Punith P

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Summary

Frontend Developer and Software Engineer who ships published developer tooling and real-time 3D web applications: five npm packages providing source-level debugging for React Three Fiber, a Fabric Minecraft mod on CurseForge and Modrinth carrying a server-authoritative security redesign, a WebGL astronomical simulator with procedural skies and geolocation-synced star fields, and a MERN-stack medical records platform with role-based access control and full audit logging. Directs AI-assisted development end-to-end — system architecture, implementation, and debugging — across all projects.

Education

Bachelor of Engineering (B.E.) in Computer Science and Engineering Sep 2023 – Jun 2027
K S Institute of Technology, Bengaluru

Skills

- **Languages:** Java, JavaScript (ES6+), TypeScript, HTML5, CSS3
- **Frontend:** React, React Three Fiber, Zustand, Tailwind CSS, Responsive Design
- **Backend:** Node.js, Express.js, REST APIs, JWT Authentication, RBAC, MongoDB, Mongoose, Multer
- **Graphics:** Three.js, WebGL, GLSL, Procedural Generation, Blender
- **Tooling & Build:** Vite, Vite Plugin Development, Babel AST Transforms, npm Workspaces, npm Package Publishing, Model Context Protocol (MCP), Git, GitHub, GitHub Actions, Gradle, Fabric API

Experience

Click-to-Source 3D — Developer Tooling for React Three Fiber npm | GitHub
TypeScript | *React Three Fiber* | *Vite Plugin* | *Babel AST* | *MCP*

- Published five interdependent npm packages (core, overlay, Vite plugin, MCP server, shared) under the @click-to-source-3d scope, MIT licensed, resolving any clicked 3D mesh to the source file, function, line, and runtime arguments that generated it
- Solved per-instance provenance for InstancedMesh across six controlled experiments by composing a build-time JSX source-location transform with runtime matrix-write interception; verified 400/400 and 805/805 instances against ground truth
- Shipped source write-back that rewrites the matched literal and hot-reloads the running scene through Vite HMR, an 8-tool MCP server exposing the same provenance to AI coding agents, and a standalone Vite plugin package keeping the core browser-pure

Lipi — Fabric Minecraft Mod CurseForge | GitHub
Java | *Fabric API* | *Custom Client-Server Networking*

- Redesigned identity verification to be fully server-authoritative across three published releases, closing a client-trust security flaw
- Built a dedicated in-game chat UI with channel sidebar, localized timestamps, unread tracking, and command autosuggestions
- Rolled out server-side moderation: persistent logging, auto-expiring mutes, per-player cooldowns, and GDPR-conscious data handling

JeevanLocker — Medical Document Management GitHub
MERN Stack | *TypeScript* | *JWT* | *RBAC*

- Led a 4-person team as Team Lead on an IEI-sponsored Student Research Project Proposal (SRPP), directing architecture and implementation
- Architected JWT authentication, bcrypt hashing, and role-based access control (Patient/Doctor/Admin) across the platform
- Built a Multer document upload and sharing pipeline with server-side permission checks — doctors access only explicitly-shared documents — plus audit logging across five event types

Low-Poly Time Sync Environment Live Demo | GitHub
Three.js | *WebGL* | *GLSL* | *SunCalc*

- Engineered a GPU-tiered, rejection-sampled star field rendering up to 7,200 stars from an 18,000-candidate pool, driven by live geolocation and real sun/moon positioning
- Developed five custom GLSL shaders (sky dome, moon-phase, star field, dust, volumetric haze) plus FBM procedural clouds for continuous day-night transitions; deployed as a zero-build ES-module app with adaptive GPU detection

Procedural Low-Poly World Generator GitHub
React Three Fiber | *Three.js* | *Procedural Generation*

- Constructed a seed-based terrain system with three archetypes, multi-layer domain warping, and instanced vegetation placed by slope and altitude
- Diagnosed a main-thread freeze to a synchronous 16,641-vertex noise computation and scoped a Web Worker offload fix; used as the dogfooding consumer that validated Click-to-Source 3D end to end

Additional Information

- **Achievements & Activities:** JeevanLocker approved for the Institution of Engineers (India) SRPP 2025–26; TCS Tech Bytes statewide engineering IT quiz participant (600+ participants); IEEE ImpactX Hackathon; Generative AI Workshop
- **Certifications:** AI Skills Passport (EY & Microsoft), AWS Cloud Practitioner (Infosys Springboard), IBM SkillsBuild Blockchain & Cybersecurity